

For the CG located at the wing quarter chord, the wing and the tail are mounted so that in level flight with an airspeed of 200 mph at standard sea level, trim is attained with no elevator deflection and a zero angle of attack for the fuselage reference line. Neglecting any effects associated with the propeller and fuselage, determine the mounting angles for the wing and the tail and the pitch stability derivative for the wing-tail combination. Use the simplified analysis presented in Sec. 4.3 and neglect the effects of wing downwash on the tail. In this and the following problems that deal with the Spitfire, any dimensions that are not otherwise given may be scaled from Figs. 4.12.1 and 4.12.2.

- 4.4. Excluding the effects of wing downwash, determine the static margin in the range of linear lift for the wing-tail combination of the Spitfire Mark XVIII that is described in problem 4.3.
- 4.5. Using the method that was presented in Sec. 4.5, determine the relation between the downwash angle on the horizontal tail and the lift coefficient for the wing of the Spitfire Mark XVIII that is described in problem 4.3.
- 4.6. Using the results that were obtained from problem 4.5 for the downwash angle, recompute the mounting angles for both the wing and tail of the Spitfire Mark XVIII that is described in problem 4.3. Compare the result to that obtained in problem 4.3.
- 4.7. Including the effects of wing downwash but neglecting any effects associated with the propeller and fuselage, determine the static margin in the range of linear lift for the wing-tail combination of the Spitfire Mark XVIII that is described in problem 4.3. Compare the result to that obtained in problem 4.4.
- 4.8. Assume that the wing and the horizontal tail of the Spitfire Mark XVIII have the following properties:

$$\begin{aligned}
 S_w &= 244 \text{ ft}^2, \quad b_w = 36.83 \text{ ft}, \quad C_{L_w, \alpha} = 4.62, \quad C_{m_w} = -0.053, \\
 \alpha_{L0w} &= -2.20^\circ, \quad \alpha_{0w} = 2.0^\circ, \quad l_w = 0.0 \text{ ft}, \\
 S_h &= 31 \text{ ft}^2, \quad b_h = 10.64 \text{ ft}, \quad C_{L_h, \alpha} = 4.06, \quad C_{m_h 0} = 0.000, \\
 \alpha_{L0h} &= 0.0^\circ, \quad \alpha_{0h} = 0.0^\circ, \quad l_h = 18.16 \text{ ft}, \\
 W &= 8,375 \text{ lbf}, \quad \varepsilon_e = 0.60, \quad C_{m_h, \delta_e} = -0.55
 \end{aligned}$$

Including the effects of wing downwash but neglecting the effects of the propeller and fuselage, find the elevator deflection and the angle of attack required for trimmed level flight at sea level for airspeeds of 100, 200, and 300 mph. Also determine the lift on the horizontal tail at level trim for the same airspeeds. Assume a tail efficiency of 1.0 and use the simplified analysis that was presented in Sec. 4.3.